

Assignment 1A

Christian O'Connor
(Dated: December 4, 2025)

INTRODUCTION

For this assignment, a finite-difference method for differential equation approximation was constructed with the help of generative AI. As an extension, error analysis was conducted after the code was working as expected.

HOW THE CODE SOLVES THE PROBLEM SET

The allowed energy levels for the quantum harmonic oscillator were equally spaced according to equation 1[1].

$$E_n = 2n + 1, \quad n = 0, 1, 2, \dots \quad (1)$$

Where n is a quantum number labelling energy level.

The eigenfunctions are shown in figure 1, stacked by their corresponding eigenvalues. The widths of these functions match the parabolic potential curve overlaid on the graph in reference to equation 1.7. The

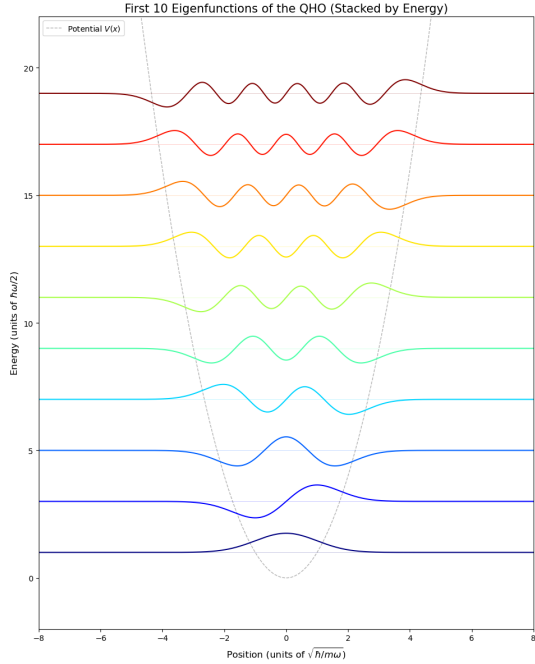


FIG. 1. A graph of the eigenfunctions of each energy level of the quantum harmonic oscillator plotted at their respective energies against position. The potential function (equation 1.7) is shown as a grey dashed line.

TABLE I. Comparison of Numerical Eigenvalues to Analytic Solutions for the Quantum Harmonic Oscillator (QHO) for a 1000-point grid.

Energy level (n)	Numeric E	Analytic E	Absolute Error
0	0.999	1.000	1.60×10^{-5}
1	2.999	3.000	8.02×10^{-5}
2	4.999	5.000	2.08×10^{-4}
3	6.999	7.000	4.01×10^{-4}
4	8.999	9.000	6.57×10^{-4}
5	10.999	11.000	9.78×10^{-4}
6	12.998	13.000	1.36×10^{-3}
7	14.998	15.000	1.81×10^{-3}
8	16.997	17.000	2.32×10^{-3}
9	18.997	19.000	2.90×10^{-3}

harmonic oscillator function produced the first n energy level values, which were compared to the analytic energy. Errors calculated thereafter were very small, confirming that this finite difference method was successfully calculating the n lowest energy levels. The numeric and analytic energies, along with the associated absolute error and respective energy level, are shown in table I. These results used an n_{points} of 1000.

The code is designed such that n can be changed easily, but for the results discussed here n was set at $n = 10$ as this value was suggested in the assignment brief.

For the Hydrogen radial solver, values of energy were compared to numeric energies defined in equation 2[2].

$$E = -\frac{1}{2n^2}, \quad n = 0, 1, 2, \dots \quad (2)$$

Table II shows the energies for the first 10 energy levels of a Hydrogen atom, if counting by quantum numbers n and ℓ . Given that the analytic energy in equation 2 depends only on n , degeneracy of eigenstates is expected. This is a well-known result for the Hydrogen atom, and is reflected in the repeated energy eigenvalues shown for the first 10 orbitals[3].

The eigenfunctions for each energy were plotted on a graph of energy against r as shown in figure 2. Accidental degeneracy can be seen in this graph, though the shape of each radial changes with ℓ , as would be expected for electrons orbiting a Hydrogen atom with different orbital angular momenta.

The results shown are as expected, given the close match to theoretical energy level values found

TABLE II. Comparison of Numerical Eigenvalues to Analytic Solutions for the quantum harmonic oscillator for a 1000-point grid.

Orbital	n	ℓ	Numeric E	Analytic E	Absolute Error
1s	1	0	-0.999923	-1.000000	7.66×10^{-5}
2s	2	0	-0.250002	-0.250000	1.60×10^{-6}
2p	2	1	-0.249995	-0.250000	4.79×10^{-6}
3s	3	0	-0.111112	-0.111111	7.36×10^{-7}
3p	3	1	-0.111111	-0.111111	6.30×10^{-8}
3d	3	2	-0.111110	-0.111111	9.46×10^{-7}
4s	4	0	-0.062500	-0.062500	3.07×10^{-7}
4p	4	1	-0.062500	-0.062500	2.84×10^{-8}
4d	4	2	-0.062500	-0.062500	1.89×10^{-9}
4f	4	3	-0.062500	-0.062500	4.05×10^{-7}

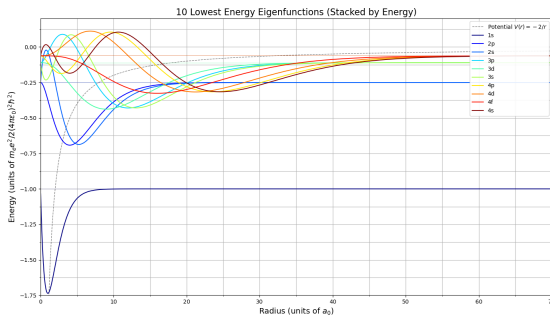


FIG. 2. A graph of respective eigenfunctions for each energy level of the Hydrogen atom plotted against radius r . The potential as defined in equation 1.11 is shown in grey.

through research and reference to the assignment brief.

The validation of individual eigenfunctions for the quantum harmonic oscillator was cross-checked by plotting each individual eigenfunction $\psi(x)$, along with its probability density function $|\psi(x)|^2$, for each energy level calculated. For the Hydrogen radial solutions, plots of wave function amplitude $u(r)$ and probability density $|u(r)|^2$ were similarly plotted for each of the 10 lowest energies. These graphs helped visualise the solutions to each problem, checking they matched theoretical descriptions of how quantum harmonic oscillators and radial wave functions should look.

METHODS

The code uses a finite-difference method to approximate the time independent Schrödinger equation (equation 1.2). This was implied by the assignment brief's allusion to discretising the spatial dimension into N points. A finite-difference solver does this in order to approximate solutions to differential equations such as equation 1.2 using finite differences. The size of these differences was varied during error

analysis, as a smaller difference corresponds to a higher resolution grid and lower errors.

The matrix representing the Hamiltonian H was broken down into two components, $H = D + V$, where D is the tri-diagonal kinetic energy matrix derived from the second-order finite-difference approximation, inferred from the brief, and V is a diagonal matrix containing the value of the potential term at each grid point such that $V_{i,i} = V(x_i)$. The relevant eigenvalues and eigenvectors were then computed using the `numpy.linalg.eigh` function because of its applicability to large, real, symmetric matrices.

The value of N was changed to find a comfortable balance between accuracy and run-time, since the method of solving multiple equations for each grid point increases run-time significantly when the grid's resolution is increased.

For the Hydrogen radial solver, r_{max} was chosen such that each radial had decayed to a constant value before the end of the graph. This value was set to 70, although lower values could have been used since all radials reach a constant value by 60. The larger value was chosen to be certain that reflection of the wave function off of boundaries would not occur, thus preventing distortion of the graph due to artefacts produced by the system's boundary conditions. This is one of the known limitations of a finite-difference solver[4].

Zeroed Boundary Conditions for the Hydrogen radial solver, i.e. $u(0) = 0$ - where hydrogen radials are solved for $r > 0$ - were used to prevent an error when plotting the energy eigenfunctions at $r = 0$ due to the r denominators in equation 1.11.

EXTENSION: ERROR ANALYSIS

As an extension, the error magnitudes for each of the first 10 energy levels was plotted against different values of N , the grid size. This required the quantum harmonic oscillator to be solved for each different value of N . Figure 3 shows the graph produced for the N range 50 – 10,000 in steps of 100 for the quantum harmonic oscillator.

The log-log style was chosen after observing the output graph using standard axes and researching the relationship of absolute error with grid resolution for finite-difference methods[5].

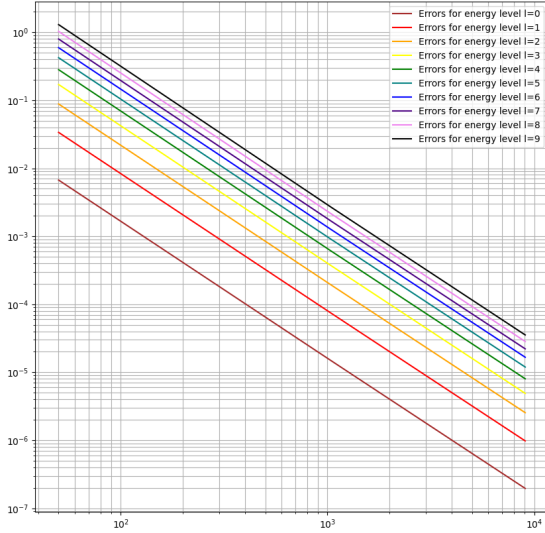


FIG. 3. A log-log graph of error magnitude for each of the lowest 10 energy levels calculated by the quantum harmonic oscillator finite-difference solver against the number of grid points N .

A similar method was attempted to display the inverse power-law relationship between absolute error in Hydrogen radial eigenvalues and N , but this method was not successfully completed within the time frame of the assignment.

CONCLUSION

The takeaways from this assignment are that the finite difference method is very good at approximating solutions to the time-independent Schrödinger equation to high precision. Specifically, the kinetic energy term of the Hamiltonian operator is accurately modelled by a tri-diagonal kinetic matrix. Imposing boundary conditions for the Hydrogen radial solver was shown to be a necessary adjustment. The reusability of the kinetic matrix function was confirmed across Cartesian and polar coordinates, and the degeneracy of orbital states for a hydrogen atom was shown by varying ℓ for each value of n used to calculate the respective energy levels.

From a quantitative assessment of the results of this assignment, it was concluded that the absolute error in the energy eigenvalues of a quantum harmonic oscillator increases as the energy level number increases. This was due to the fixed grid size Δx failing to resolve the denser oscillations observed at higher values of n , as seen in figure 1. This is known as discretisation error and is inherent to the finite difference method[6].

From the extension conducted, the constant negative gradient of figure 3 allows us to infer that the number

of grid points N and the absolute error in energy have an inverse power-law relationship, confirming the expected second-order accuracy of the finite-difference method and showing the error is proportional to $(\Delta x)^2$, the spacing between grid points.

Further experimentation could involve investigating the balance between error and resolution for Hydrogen radial solutions and higher energy levels. Since higher grid resolution improves accuracy for denser high- n waveforms but significantly increases run-time, determining an optimal balance could be useful for application to other physical systems. Parallel processing could be used to minimise the computational cost.

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USE OF GENERATIVE AI

The AI tools used were: ChatGPT (free), Google Gemini (pro) and Claude (free). The main uses of AI were to generate code given the assignment brief, then to cross-check the code with other AI models, determining if it was answering the problem correctly or not. Code was either copied from the AI model outputs in entirety or in snippets, but code was often changed manually after consulting responses from AI models.

The prompts used are mostly described in markdown cells in the Jupyter Notebook, but some prompts are written in comments where most relevant.